Kushagra Kaushal

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Summary

Member of Technical Staff with over three years of experience across agentic AI systems, cloud-native infrastructure, data engineering, machine learning, analytics, and applied research. Experienced in building production AI platforms, agent harnesses, tool-calling systems, MCP/plugin integrations, Cloudflare-based runtimes, scalable data pipelines, and client-facing product workflows. Strong focus on turning LLM-based workflows into reliable, observable, and user-facing products.

Education

Indian Institute of Information Technology — Bhopal (India)
B.Tech Information Technology

Skills

- **Programming Languages:** Python, TypeScript, JavaScript, C++, C
- **AI/ML:** LLM Agents, Agent Harnesses, Tool Calling, MCP, RAG/GraphRAG, Model Routing, Scikit-learn, TensorFlow, PyTorch, Computer Vision, NLP
- **Agent Infrastructure:** Cloudflare Workers, Cloudflare Sandboxes, Cloudflare AI Gateway, Isolated Execution, Skills/Scripts Management
- **Data Engineering Technologies:** dbt, Apache Airflow, Google BigQuery, Cloud SQL
- **Cloud Platforms:** Google Cloud Platform, Amazon Web Services
- **Data Visualization & Reporting Tools:** Looker Studio, Plotly, Dash
- **DevOps & MLOps Technologies:** Kubernetes, Docker, Devtron CI/CD
- **Additional Technologies:** Linux, MySQL, Bash Scripting, Node.js, FastAPI, Flask, Django, REST APIs, OAuth, Webhooks

Work Experience

Member of Technical Staff Zonko Labs - Mumbai, India (Onsite) — April 2026 - Present

- Building Harbor, an AI workbench and integration platform that lets agents access tools across MCP, APIs, and CLIs through a unified plugin architecture and execution layer.
- Designed Harbor's CodeMode execution layer using JavaScript/TypeScript-style tool calling, enabling tools to be composed, piped, and reused reliably across complex multi-step agent tasks.
- Developed Harbor's runtime and plugin system on Cloudflare Workers and AI Gateway, covering tool discovery, OAuth-based connections, secrets/policy handling, model routing, skills, and reusable scripts/workflows.
- Building Luffy, an AI coworker for knowledge-work teams, with workspace context, skills/scripts management, persistent execution flows, and Harbor-backed integrations for tool use.
- Architected Luffy runtime paths on Cloudflare Workers and Sandboxes for scalable request handling, isolated sessions, and ad-hoc or compute-heavy agent workloads.
- Performed FDE work by meeting clients, understanding use cases and feedback, translating requirements into integration scope, and shaping the Harbor/Luffy product roadmap.

Data Scientist New Engen - Seattle, US (Remote) — November 2022 - March 2026
`https://www.newengen.com/`

- Architected Lift AI, a real-time distributed chatbot platform with multi-agent reasoning, tool calling (MCP), and graphRAG-based retrieval, designed for millions of concurrent users with task-based distributed processing and resilient system design.
- Built scalable data infrastructure integrating Adverity, dbt, BigQuery (OLAP), and Postgres on CloudSQL (OLTP), implementing hybrid ETL/ELT pipelines that unified 5TB+ of cross-platform marketing data.
- Developed OLAP pipelines using Airflow and BigQuery for real-time and scheduled analytics, enabling high-throughput data science and reporting workloads.
- Delivered production-grade models such as marketing mix modeling and revenue forecasting, driving budget optimization and ROI improvements through automated, data-driven insights.

Data Science Consultant Kauriink Pvt. Ltd. - New Delhi, India (Remote) — August 2022 - November 2022
`https://www.techatplay.ai/`

- Developed and validated deep learning models for automated player performance analysis using computer vision, achieving 87% accuracy in posture classification (6 classes) and 92% accuracy in shot type classification (8 classes).
- Implemented advanced color segmentation and sliding window techniques to optimize object tracking in video data.
- Collaborated on the integration of transfer learning models (Mediapipe, YOLO) to accurately detect player posture, movement, ball trajectory, and shot type.
- Exhibited exceptional problem-solving and analytical capabilities, applying technical expertise and innovative approaches to deliver high-impact solutions.

Research Experience

Research Assistant University of Michigan - Michigan, US (Remote) May 2023 - Dec 2023

- Designed an encoder-decoder neural network architecture to generate high-fidelity simulated LiDAR point cloud data from 3D scenes.
- Leveraged Blender and Python automation for efficient dataset creation, incorporating advanced object translation and scaling techniques.
- Engineered a comprehensive model pipeline, encompassing data pre-processing, PointNet-based encoder, decoder, and a tailored loss function utilizing a modified Chamfer Distance metric.

Research Intern IIT Indore - Indore, India (Remote) May 2022 - February 2023

- Developed advanced machine learning models to predict speaker attributes (age, height, sex) from audio recordings with high accuracy.
- Extracted and normalized acoustic features (MFCC, Mel-filterbank) using CMVN techniques to enhance model performance.
- Applied sophisticated audio data augmentation methods, including speed perturbation and masking, to improve model robustness.
- Designed and trained models leveraging transformers, Wav2Vec, and SincNet, achieving performance comparable to or surpassing state-of-the-art benchmarks.

Projects

Obelisk Ongoing

`https://github.com/kshgrk/obelisk`

- Developed Obelisk, a real-time chat application leveraging Temporal workflows for fault-tolerant orchestration and OpenRouter's free AI models (e.g., DeepSeek, Mistral, Llama), enabling dynamic model switching mid-conversation without context loss.
- Architected a multi-layered system using FastAPI for proxy and backend services, Vanilla JavaScript frontend, and SQLite database, facilitating seamless session persistence, real-time streaming responses, and live updates.
- Implemented an extensible tool registry for dynamic tool calling, including built-in integrations for calculators and weather APIs, with a plugin-based architecture allowing easy addition of custom tools at runtime.
- Ensured robustness through Temporal's retry mechanisms and layered design, supporting resumable conversations, model-agnostic sessions, and integration with external services for enhanced functionality.

IntelliCodebase Jan 2025

`https://github.com/kshgrk/IntelliCodebase.git`

- Architected an advanced LLM-powered system (utilizing Gemini) for comprehensive codebase analysis and modification.
- Integrated function calling with Bash and Python to execute system-level tasks, including file creation, deletion, and script execution.
- Implemented an optimized content caching mechanism to enhance performance during analysis of large-scale codebases.
- Developed functionality to identify codebase issues and propose targeted fixes, with selective file analysis capabilities.

LSMTree-AVL Nov 2024

`https://github.com/kshgrk/LSMTree-AVL.git`

- Engineered a Python-based LSM-tree database utilizing AVL trees for efficient memtable and index management, optimizing storage and retrieval operations.
- Incorporated a Bloom filter for rapid key presence verification, memtable flushing to SSTables, and multi-threaded compaction strategies.
- Developed a write-ahead log mechanism to ensure data durability and robust recovery processes.